Day 29

# Encryption

* Now we will discuss about encryption and Decryption mechanism.

→ usually we will do Integrating the Interfaces that with multiple Systems, ∞

eg bank

* What ever the data you are sending to the bank it is sensite data → because it Contain's your personal Information, So when you are integrating with Sensitive Information data → how can you make sure the data is secure ?

→ how can you do message level security

We can youse encoder right? → yes But the problem is → encoder will just encode the data → and any body can go and decode the data → right?

So Instead we have one more option called Encryption, So you can encrypt the data and no body can decrypt the data

* When you are encrypting the data you will encrypt the data by using a [key] → you will decrypt the data by using particular key

eg:- Encryption → message level encryption

So When discussing about SFTP Connection we discussed about public key authentication right

Now → when you are creating public key based Authentication → ① create SSH key

↓

certificate Based Authentication

public key + private key

↓

Combination of certificates

Both are part of certificate

↓

That is the reason we will call it as key pair Certificate

2 certificates (Key pair).

1. Public key
2. private key

As per the name Suggest's → public key

We can Share this public key with anyone

Private: This the one which Cannot be shared it's our personal key

⇒ When you are encrypting the data you will use these keys and also for decrypting the data

So, let's Say we are locking the room, right → Now if you want to unlock that room → what is the requirement →

∴ The key which is used to lock it → that Same key is used to unlock it

* let's cp1 certificate → cp1 public key
cp1 private key

3rd party Public key
3rd party private key -

* So above keys are interlinked → how?

Suppose if the message data is Sensitive. and it is encrypted use CPI public key then you need CPI private key to decrypt it and. We cannot use 3rd party private key because they are Inter linked.

* It's your responsibility to Secure the Sensitive message Information using encryption techniques

⇒ In CPI we have one encryption mechanism called PGP Encryption Mechanism

PGP → Pretty Good print → Most of the Organizations will use this

This one also have two types of keys

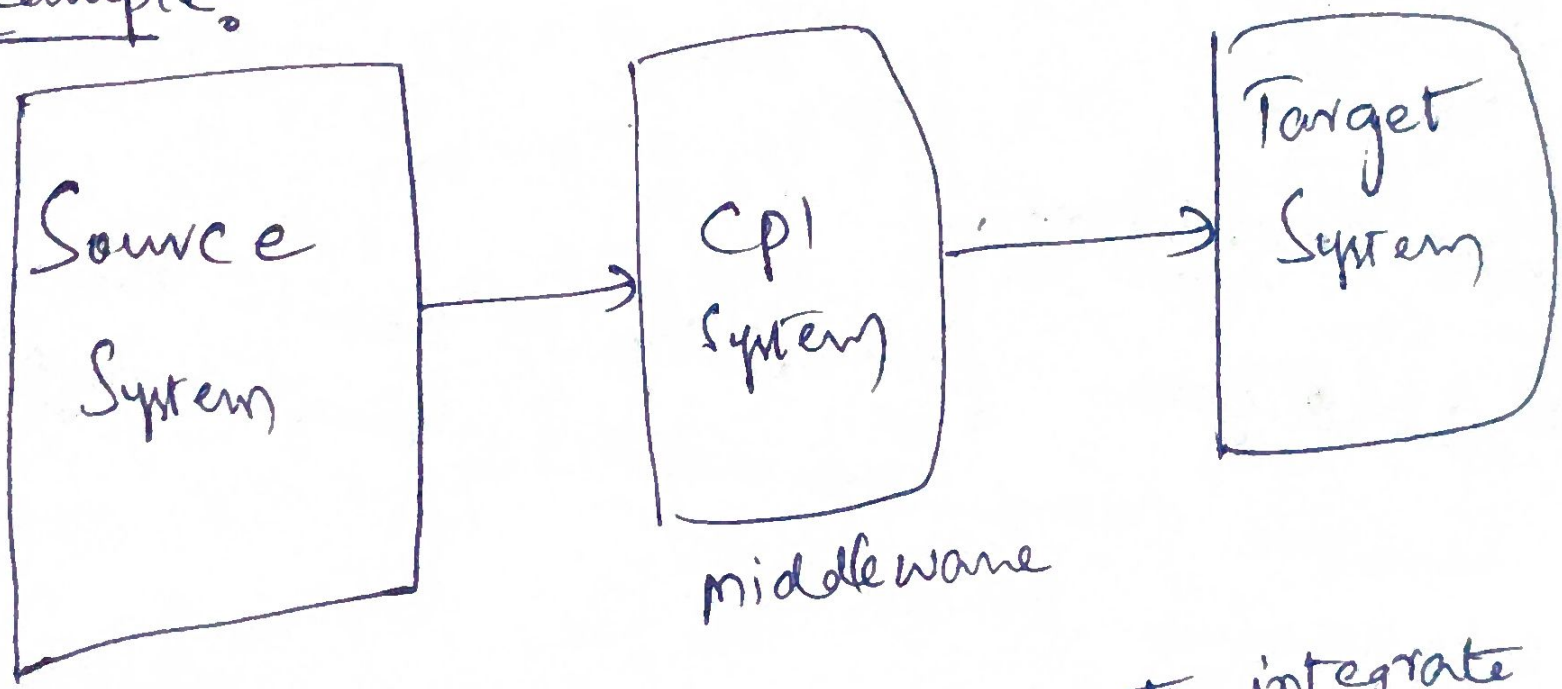
PGP public key -
PGP private key -

* When ever you are encrypting the data you will encrypt with private key

Encrypting ---> Public key

Decrypting ⟶ private key

Example:




So, here we are using CPI to integrate between Source System and target System

So here the requirement is → what ever the payload? You are sending, that data,

(m)
Body
(op)

Message Content

Should be encrypted

* So while encrypting that we need to use

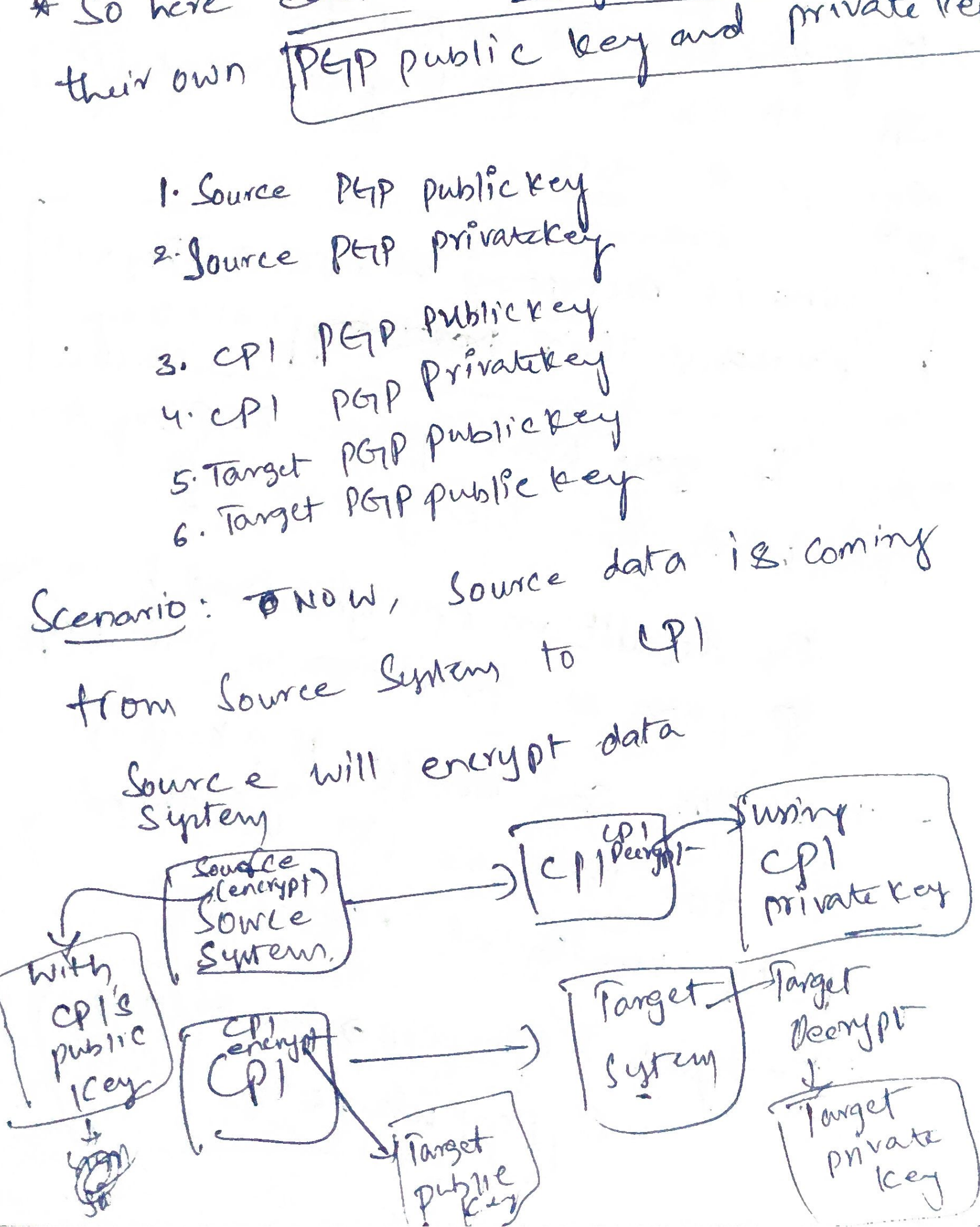
**PGP public key and private key**

* So here each and every system will have their own **PGP public key and private key**

1. Source PGP publickey
2. Source PGP privatekey
3. CP1 PGP publickey
4. CP1 PGP Privatekey
5. Target PGP publickey
6. Target PGP public key

Scenario: NOW, Source data is coming from Source System to CP1

Source will encrypt data



* This is nothing But encryption and decryption.

* So you need understand with public key you need to encrypt the data and with which private key you need to decrypt the data

***

∴ Who is decrypting the data → They will exchange their ~~private key~~ public key.

* Now, if you want generate PGP key for CPI where can we generate it?

By default in CPI you Don't have the option of generation of PGP keys

→ then how can we get PGP keys → So for that we will use third party tool called "openPGP"

→ google → openPGP → downloaded → GPG4win.exe

Kleopatra → Gpg4win 5.10

* In real time Scenario's also, we use this.

Software:

→ keymaterial → rsa2048

Algorithm

↓

public k's generated → export → done

to local System

→ for private key → Backup secret key

→ Now we need to import this Both key in CP System → how we will do that?

⇒ Monitor tab → Mange Security → PGP keys

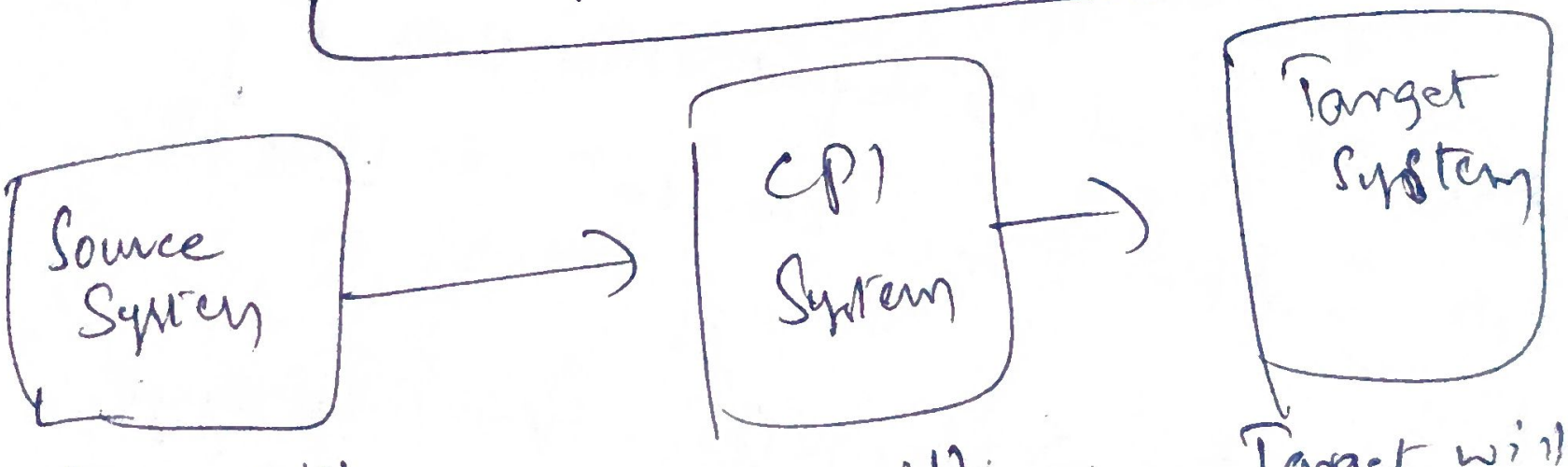
Add → public key

Secret key

* In Some Cases with encryption, they will ask to Sign the payload (Signature). So that we will verify that payload is verified payload or not

→ While decryption they will verify that Signature also ✓

*** Signing ---→ PGP private key
Verify ---→ PGP public key




Source will encrypt → CPI public key & Sign the data Source → private key ✓

CPI will decrypt → CPI private key & Sign the data CPI private key & verify the data using Source Public key

CPI will encrypt → Target Public key

Target will decrypt → CPI Public key & verify the data Target private key

CPI public key ✓

* When you are encrypting the data you need to check whether we need to do signing or not if they say yes → then you have to ask for their public key also

↓

this one will be depends on the requirement

PGP Encrypter : processing :

Signatures.

Content Encryption Algorithm.

Secret key Algorithm.

Compression Algorithm.

Armored :

Integrity Protected Data packet.

Encryption key user ID : public key

Signature → Sign the data using

Private key

PGP Decrypter : processing

Signatures: Required

↓

It will verify the data using public key

* PKCS. Encrypter:

public key → usage

* PKCS Decrypter :-

download certificate

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POC → Two poc's

① CPI → push (place) files.com → } But

② CPI → pull files.com →

encrypt it (here we will understand SFTP server will decrypt or not)

pull encrypted file from from SFTP → decrypt in CPI